Aleksas Mazeliauskas

Theoretical Physics Department
CERN Building 4/2-065
CH-1211 Geneva 23
Switzerland

Email aleksas.mazeliauskas@cern.ch
Phone +41 22 76 73204
Website aleksas.eu
Nationality lithuanian

Professional experience

2019–2021 Senior research fellow, **Theoretical Physics Department, CERN**

Research on heavy-ion phenomenology, QCD thermalization and future ion experiments at LHC.

2017–2019 Postdoctoral researcher, **Institute for Theoretical Physics, Heidelberg University**

Member of collaborative research center SFB 1225 *Isolated quantum systems and universality in extreme conditions*.

Work on non-equilibrium QCD physics using quark and gluon kinetic theory, pre-equilibrium kinetic propagator `KøMPøST`, resonance decay code `FastReso`.

2014–2017 Research assistant, **Nuclear Theory Group, Stony Brook University**

PhD dissertation work under supervision of Prof. Derek Teaney, *Fluctuations in ultra-relativistic heavy ion collisions*

Education

2012–2017 **Stony Brook University**, United States, Doctor of Philosophy in Physics
Max Dresden Prize for outstanding theoretical thesis, degree awarded on 19/05/2017

2011–2012 **University of Cambridge**, United Kingdom, Master of Mathematics (Part III Maths)
graduated with distinction

2008–2011 **University of Cambridge**, United Kingdom, Bachelor of Arts (Mathematical Tripos)
1st class degree

Awards

09/11/2019 Nuclear Physics A Young Scientist Award for the best theory talk at Quark Matter 2019

02/05/2017 Max Dresden Prize for outstanding theoretical thesis

25/07/2016 APS FGSA Travel Award for Excellence in Graduate Research

23/05/2013 David Fox award for the outstanding Teaching Assistant

Professional services

Events organised

2020 Co-organiser of **Baltic School of High-Energy Physics and Accelerator Technologies**, 10-14 Aug 2020, Klapkalnciems, Latvia. 20k€ funding acquired from COST Action CA16201.

2019 Convener of discussion group on "Connections between QGP-like observables in small systems" at **International Workshop on QCD Challenges from pp to AA**, 19-23 Aug 2019, Lund, Sweden, and the editor of the summary report.

Convener of discussion group on "Implementations of critical dynamics" at **EMMI Rapid Reaction Task Force: Dynamics of critical fluctuations**, 8-12 April 2019, Darmstadt, Germany, and the editor of the summary report.

Refereeing

Physical Review Letters, Physical Review C and D, Physics Letters B, Nuclear Physics B

Languages

Lithuanian	native, enthusiastic teacher	**French**	elementary (A2)
English	fluent, higher education in English	**Russian**	basic conversational
German	upper intermediate (B2)	**Chinese**	beginner, passed HSK level II test

Programming

Programming	Extensive numerically oriented coding in `C++` and `Python`, previous experience with `Fortran` and Matlab, good working knowledge of Mathematica
General IT	Git (source code management), LaTeX (word processing), Inkscape (vector graphics), GNU/Linux (operating system), Gnuplot (graphing), HTML5 (web building)

Invited presentations

28/05/2020 talk at **VIII Large Hadron Collider Physics Conference (online)** on *Equilibration of QGP in small systems*, Paris, France

06/12/2019 talk at **XIX Zimányi School Winter Workshop** on *Pre-equilibrium phenomena in Quark Gluon Plasma*, Budapest, Hungary

25/11/2019 talk at **Theoretical Foundations of Relativistic Hydrodynamics Workshop** on *Pre-scaling, hydrodynamic attractors and entropy production in heavy ion collisions*, Banff, Canada

01/11/2019 talk at **New Development of Hydrodynamics and its applications in Heavy-ion Collisions Workshop** on *Hydrodynamic attractors, initial state energy and particle production in relativistic nuclear collisions*, Shanghai, China

13/09/2019 talk at **XLIX International Symposium on Multiparticle Dynamics** on *Equilibration in Quark Gluon Plasma*, Santa Fe, New Mexico, USA

25/06/2019 plenary talk at **V Initial Stages Conference** on *Matching of initial conditions to hydro evolution*, New York, USA

06/04/2019 talk at **XIV Polish Workshop on Relativistic Heavy-Ion Collisions** on *Chemical equilibration in hadronic collisions*, Krakow, Poland

01/08/2018 parallel talk at **XIII Quark Confinement and the Hadron Spectrum Conference**, *2+1D simulations of pre-equilibrium stage with QCD kinetic theory*, Dublin, Ireland

18/05/2018 plenary talk at **XXVII Quark Matter Conference** on *Initial conditions for nuclear collisions: theory overview*, Venice, Italy

02/10/2017 talk at **XII High-pT Physics Workshop** on *Initial conditions for heavy ion collisions* , Bergen, Norway

15/06/2017 talk at **XII Workshop on Particle Correlations and Femtoscopy** on *Kinetic theory equilibration for realistic heavy ion initial conditions*, Amsterdam, The Netherlands

06/03/2015 talk at **Collectivity in Small Colliding Systems with High Multiplicity Workshop** on *Principal Component Analysis and Subleading Flow*, Brookhaven National Lab, New York, United States

Contributed presentations

03/06/2020 **X Hard Probes (online)**, Austin, USA

06/11/2019 **XXVIII Quark Matter Conference**, Wuhan, China, BEST THEORY TALK

22/09/2019 **Quantum Systems in Extreme Conditions**, Heidelberg, Germany

10/06/2019 **XVIII Strangness in Quark Matter**, Bari, Italy

14/01/2019 **XLVII Workshop on Gross Properties of Nuclei and Nuclear Excitations**, Hirschegg, Austria

23/05/2019 **Origins of Correlations in High Energy Collisions**, INT, Seattle, Washington, United States

09/04/2019 **EMMI RRTF: Dynamics of critical fluctuations**, Darmstadt, Germany

02/10/2018 **IX Hard Probes conference**, Aix-Les-Bains, France

27/08/2018 **Probing the Quark-Gluon Plasma with collective phenomena and heavy quarks**, MIAPP, Munich, Germany

19/09/2017 **IV Initial Stages**, Krakow, Poland

08/08/2017 **Critical Point and Onset of Deconfinement**, Stony Brook, New York, United States

07/02/2017 **XXVI Quark Matter**, Chicago, Illinois, United States

19/09/2016 **VII Hot Quarks**, South Padre Island, Texas, United States

25/05/2016 **III Initial Stages**, Lisbon, Portugal

22/07/2015 **Correlations and Fluctuations in p+A and A+A Collisions**, INT, Seattle, Washington, United States

27/03/2015 **Ohio-Region APS meeting**, Kent, Ohio, United States

Publications

2020

1. J. Berges, M.P. Heller, A. Mazeliauskas, R. Venugopalan, *Thermalization in QCD: theoretical approaches, phenomenological applications, and interdisciplinary connections*, arXiv:2005:12299
2. EDITOR FOR J. Adolfsson et al., *QCD Challenges from pp to A-A Collisions*, community report, arXiv:2003.10997
3. EDITOR FOR M. Bluhm et al., *Dynamics of critical fluctuations: Theory – phenomenology – heavy-ion collisions*, community report, arXiv:2001.08831

2019

4. O. Garcia-Montero, N. Löher, A. Mazeliauskas, J. Berges, and K. Reygers, *Probing the evolution of heavy-ion collisions using direct photon interferometry*, arXiv:1909:12246
5. D. Devetak, A. Dubla, S. Floerchinger, E. Grossi, A. Mazeliauskas, S. Masciocchi, and I. Selyuzhenkov, *Global fluid fits to identified particle transverse momentum spectra from heavy-ion collisions at the Large Hadron Collider*, **JHEP 2006 (2020) 44** arXiv:1909:10485
6. P. Hanus, A. Mazeliauskas, and K. Reygers, *Entropy production in pp and Pb-Pb collisions at energies available at the CERN Large Hadron Collider*, **Phys. Rev. C 100, 064903 (2019**), arXiv:1908.02792
7. G. Giacalone, A. Mazeliauskas and S. Schlichting, *Hydrodynamic Attractors, Initial State Energy, and Particle Production in Relativistic Nuclear Collisions*, **Phys. Rev. Lett. 123, 262301 (2019)**, arXiv:1908.02866
8. A. Mazeliauskas and V. Vislavicius, *Temperature and fluid velocity on the freeze-out surface from π, K, p spectra in pp, p–Pb and Pb–Pb collisions*, **Phys. Rev. C 101, 014910 (2020)**, arXiv:1907.11059
9. W. Florkowski, A. Kumar, R. Ryblewski, and A. Mazeliauskas, *Longitudinal spin polarization in a thermal model*, **Phys. Rev. C 100, 054907 (2019)**, arXiv:1904.00002

2018

10. A. Kurkela, A. Mazeliauskas, *Chemical equilibration in weakly coupled QCD*, **Phys. Rev. D 99, 054018 (2019)**, EDITORS' SUGGESTION, arXiv:1811.03068
11. A. Kurkela, A. Mazeliauskas, *Chemical equilibration in hadronic collisions*, **Phys. Rev. Lett. 122, 142301 (2019)**, arXiv:1811.03040
12. A. Mazeliauskas, J. Berges, *Prescaling and far-from-equilibrium hydrodynamics in the quark-gluon plasma*, **Phys. Rev. Lett. 122, 122301 (2019)**, arXiv:1810.10554
13. A. Mazeliauskas, S. Floerchinger, E. Grossi, D. Teaney, *Fast resonance decays in nuclear collisions*, **Eur. Phys. J. C (2019)**, arXiv:1809.11049
14. A. Kurkela, A. Mazeliauskas, J.-F. Paquet, S. Schlichting and D. Teaney, *Matching the non-equilibrium initial stage of heavy ion collisions to hydrodynamics with QCD kinetic theory*, **Phys. Rev. Lett. 122, 122302 (2019)**, EDITORS' SUGGESTION, arXiv:1805.01604
15. A. Kurkela, A. Mazeliauskas, J.-F. Paquet, S. Schlichting and D. Teaney, *Effective kinetic description of event-by-event pre-equilibrium dynamics in high-energy heavy-ion collisions*, **Phys. Rev. C,99, 034910 (2019)**, arXiv:1805.00961

pre-2018

16. Y. Akamatsu, A. Mazeliauskas and D. Teaney, *Bulk viscosity from hydrodynamic fluctuations with relativistic hydro-kinetic theory*, **Phys. Rev. C 97, 024902 (2018)**, arXiv:1708.05657
17. Y. Akamatsu, A. Mazeliauskas and D. Teaney, *A kinetic regime of hydrodynamic fluctuations and long time tails for a Bjorken expansion*, **Phys. Rev. C,95, 014909 (2017)**, arXiv:1606.07742
18. L. Keegan, A. Kurkela, A. Mazeliauskas and D. Teaney, *Initial condition for hydrodynamics from weakly coupled pre-equilibrium evolution*, **JHEP 1608 (2016) 171**, arXiv:1605.04287
19. A. Mazeliauskas and D. Teaney, *Fluctuations of harmonic and radial flow in heavy ion collisions with principal components*, **Phys. Rev. C 93, 024913 (2016)**, arXiv:1509.07492
20. A. Mazeliauskas and D. Teaney, *Subleading harmonic flows in hydrodynamic simulations of heavy ion collisions*, **Phys. Rev. C 91, 044902 (2015**), arXiv:1501.03138

Conference proceedings

1. A. Kurkela and A. Mazeliauskas, **SQM 2019**, *Chemical equilibration of QGP in hadronic collisions*, to be published in Springer Proc. Phys., arXiv:1910.06664.
2. A. Kurkela, A. Mazeliauskas, J. Paquet, S. Schlichting and D. Teaney, **Confinement XIII**, *Matching the non-equilibrium initial stage of heavy ion collisions to hydrodynamics with QCD kinetic theory,* PoS Confinement2018 (2019), 152
3. A. Mazeliauskas, **Quark Matter 2018** *Initial conditions for nuclear collisions: theory overview*, Nucl. Phys. A 982 (2019), 134-141, arXiv:1807.05586
4. A. Kurkela, A. Mazeliauskas, J. Paquet, S. Schlichting and D. Teaney, **HPT 2017**, *Initial conditions for heavy ion collisions with QCD kinetic theory*, PoS High-pT2017 (2019), 001
5. A. Kurkela, A. Mazeliauskas, J. Paquet, S. Schlichting and D. Teaney, **HP2018**, *2+1D simulations of pre-equilibrium stage with QCD kinetic theory*, PoS HardProbes2018 (2018), 111
6. A. Mazeliauskas, Y. Akamatsu and D. Teaney, **CPOD 2017**, *Out-of-equilibrium hydrodynamic fluctuations in the expanding QGP*, PoS CPOD2017 (2018), 038

7. Y. Akamatsu, A. Mazeliauskas and D. Teaney, **Quark Matter 2017**, *A kinetic regime of hydrodynamic fluctuations and long time tails for a Bjorken expansion*, Nucl. Phys. A 967 (2017), 872-875, arXiv:1705.08199 (talk presented by Y. Akamatsu)
8. A. Kurkela, A. Mazeliauskas, J. Paquet, S. Schlichting and D. Teaney, **Quark Matter 2017**, *Initial conditions for hydrodynamics from kinetic theory equilibration*, Nucl. Phys. A 967 (2017), 289-292, arXiv:1704.05242
9. A. Mazeliauskas, **HQ2016**, *Initial conditions for hydrodynamics from weakly coupled pre-equilibrium evolution*, J. Phys. Conf. Ser. 832 (2017) no.1, 012016, arXiv:1610.09912

PhD thesis

A. Mazeliauskas, *Fluctuations in ultra-relativistic heavy ion collisions*, arXiv:1702.01202, Max Dresden Prize for outstanding theoretical thesis, degree awarded on 19/05/2017